

SmartMarket — Multi-Branch E-Commerce System

Prepared By

- **Student Name:** Mahmoud Mohamed Anwer
 - **Student ID:** 230105170
 - **Instructor:** Eng. Soliman Younis
-

Table of Contents

1. Introduction
 2. Problem Statement
 3. Project Objectives
 4. System Overview
 5. Technologies Used
 6. System Architecture
 7. Database Design
 8. User Roles and Permissions
 9. Core Features
 10. Security Implementation
 11. Middleware Implementation
 12. Models and Relationships
 13. User Interface Design
 14. Project Workflow
 15. Testing and Validation
 16. Challenges and Solutions
 17. Future Enhancements
 18. Conclusion
-

1. Introduction

SmartMarket is a professional multi-branch e-commerce system developed using Laravel 12 and MySQL. The system is designed to help companies manage online sales, branch inventories, customer orders, and administrative operations through a centralized platform.

The application provides a complete shopping experience for customers while giving administrators and branch managers powerful tools for inventory control, reporting, and order management.

The project focuses on security, scalability, maintainability, and modern web development practices.

2. Problem Statement

Many retail companies operate through multiple branches, but traditional systems often fail to provide:

- Centralized inventory management
- Real-time branch stock tracking
- Secure order processing
- Role-based management systems
- Detailed sales reporting
- Efficient customer experience

SmartMarket solves these problems by creating a centralized platform where:

- Customers can order from different branches
 - Each branch manages its own inventory
 - Administrators monitor all branches
 - Orders and stock are synchronized efficiently
-

3. Project Objectives

The main objectives of SmartMarket are:

- Build a secure e-commerce platform using Laravel 12
- Support multiple store branches
- Implement inventory management per branch

- Create role-based access control
 - Develop a modern responsive user interface
 - Secure the system against common web attacks
 - Generate sales and inventory reports
 - Improve shopping experience for customers
-

4. System Overview

The system consists of three major user types:

4.1 Administrator

The administrator has full control over the system including:

- Managing products
- Managing categories
- Managing branches
- Managing users
- Monitoring orders
- Viewing reports
- Controlling inventory

4.2 Branch Manager

Each branch manager can:

- Monitor branch inventory
- View branch-related orders
- Manage stock levels
- Access branch dashboard

4.3 Customer

Customers can:

- Browse products
- Search and filter items
- Add items to cart
- Place orders
- Select branches
- Track order history
- View product reviews

5. Technologies Used

Technology	Purpose
Laravel 12	Backend Framework
PHP 8.2	Server-side programming
MySQL 8	Database management
XAMPP	Local server environment
Blade Engine	Frontend templating
HTML5/CSS3	User interface
JavaScript ES6	Interactive features
Font Awesome	Icons
Google Fonts	Typography
PHP GD	Image generation

6. System Architecture

The application follows the MVC (Model-View-Controller) architecture.

6.1 Model Layer

Responsible for database communication and relationships.

Examples:

- User Model
- Product Model
- Order Model
- Inventory Model

6.2 View Layer

Implemented using Laravel Blade templates.

Responsible for:

- User interface
- Product pages
- Dashboards
- Reports

6.3 Controller Layer

Responsible for business logic such as:

- Authentication
 - Order processing
 - Cart management
 - Product operations
-

7. Database Design

The database contains multiple related tables.

Main Tables

Users Table

Stores:

- Name
- Email
- Password
- Role
- Address
- Phone

Products Table

Stores:

- Product name
- Price
- Description
- Images
- Category

Branches Table

Stores:

- Branch name
- Location

- Manager
- Contact information

Orders Table

Stores:

- Order details
- Customer information
- Payment status
- Shipping information

Inventories Table

Stores:

- Product quantity per branch
- Low stock alerts

8. User Roles and Permissions

Admin Permissions

- Full system access
- Manage products and categories
- Manage users and branches
- View reports and statistics
- Control inventory and orders

Branch Manager Permissions

- Access branch data only
- Manage branch inventory
- View branch orders

Customer Permissions

- Browse products
- Create orders
- Manage cart
- Track orders

9. Core Features

9.1 Authentication System

The system includes:

- User registration
- Login and logout
- Remember me feature
- Session security
- Password validation

9.2 Product Management

Admins can:

- Add products
- Edit products
- Delete products
- Upload images
- Set sale prices
- Assign categories

9.3 Inventory Management

The system supports:

- Inventory per branch
- Low stock alerts
- Quantity tracking
- Stock updates

9.4 Shopping Cart

Customers can:

- Add products to cart
- Update quantities
- Remove products
- Clear cart

9.5 Checkout System

Includes:

- Shipping information
- Branch selection
- Payment method selection
- Order summary

9.6 Reporting System

Reports include:

- Revenue reports
 - Top-selling products
 - Branch performance
 - Daily sales statistics
-

10. Security Implementation

Security is one of the most important parts of the project.

10.1 SQL Injection Prevention

Laravel Eloquent ORM uses prepared statements automatically.

Example:
<code>Product::where('name', 'like', '%' . \$search . '%')->get();</code>

10.2 XSS Protection

The system sanitizes user input using:

<code>htmlspecialchars();</code> <code>strip_tags();</code>
--

Blade templates also escape outputs automatically.

10.3 CSRF Protection

Laravel CSRF tokens protect forms against forged requests.

10.4 Rate Limiting

Login attempts are limited to prevent brute-force attacks.

10.5 Secure File Uploads

The application validates:

- File type
- MIME type
- File size
- Random filenames

10.6 Security Headers

The application adds:

- X-Frame-Options
- Content-Security-Policy
- X-Content-Type-Options
- Referrer-Policy

11. Middleware Implementation

Middleware provides request filtering and security.

Middleware	Purpose
AdminMiddleware	Restricts admin pages
BranchManagerMiddleware	Restricts manager pages
SecurityHeadersMiddleware	Adds security headers
SanitizeInputMiddleware	Prevents XSS
PreventSqlInjectionMiddleware	Detects dangerous SQL patterns

12. Models and Relationships

One-to-One Relationships

- User → Cart
- Branch → Manager

One-to-Many Relationships

- Category → Products
- User → Orders
- Order → Order Items
- Product → Reviews

Many-to-Many Relationships

- Products ↔ Branches through inventories table

These relationships improve database organization and system scalability.

13. User Interface Design

The interface uses a professional dark theme.

Design Features

- Responsive design
- Dark navy background
- Orange primary color
- Mobile-friendly layout
- Professional dashboards
- Sidebar navigation

UI Technologies

- HTML5
 - CSS3
 - Blade Templates
 - Font Awesome Icons
 - Google Fonts
-

14. Project Workflow

Customer Workflow

1. Register/Login
2. Browse products
3. Add items to cart
4. Checkout
5. Select branch
6. Complete order
7. Track order history

Admin Workflow

1. Login to dashboard
 2. Manage products
 3. Monitor orders
 4. Manage inventory
 5. Generate reports
-

15. Testing and Validation

The project was tested for:

Functional Testing

- Authentication
- Product CRUD operations
- Cart functionality
- Order placement
- Inventory updates

Security Testing

- SQL Injection attempts
- XSS payload testing
- Unauthorized access testing
- Session validation

UI Testing

- Mobile responsiveness
- Navigation testing
- Form validation

16. Challenges and Solutions

Challenge	Solution
Multi-branch inventory management	Used pivot inventory table
Secure authentication	Laravel authentication system
SQL injection prevention	Eloquent ORM + middleware
Role management	Custom middleware and authorization
File upload security	MIME validation and random filenames

17. Future Enhancements

Future improvements may include:

- Online payment gateway integration
 - Real-time notifications
 - AI-based product recommendations
 - REST API for mobile applications
 - Advanced analytics dashboard
 - Barcode and QR support
 - Multi-language support
 - Live chat support
-

18. Conclusion

SmartMarket is a secure and scalable multi-branch e-commerce system developed using modern Laravel technologies.

The system successfully provides:

- Professional e-commerce functionality
- Multi-branch inventory management
- Secure authentication and authorization
- Advanced reporting features
- Responsive user interface
- Protection against common web attacks

The project demonstrates strong understanding of:

- Laravel Framework
- Database Design
- Web Security
- MVC Architecture
- Frontend and Backend Integration
- Software Engineering Principles

The system can be expanded in the future to support larger enterprise-level operations.

References

1. [Laravel Official Documentation](#)
 2. [PHP Official Documentation](#)
 3. [MySQL Documentation](#)
 4. [OWASP Web Security Guidelines](#)
 5. [Font Awesome Documentation](#)
 6. [MDN Web Docs](#)
-